

Lab Sheet 6: MongoDB Basic commands

Branch/ Class: B.Tech/M.Tech

Date:

Faculty Name: Prof. S.Gopikrishnan

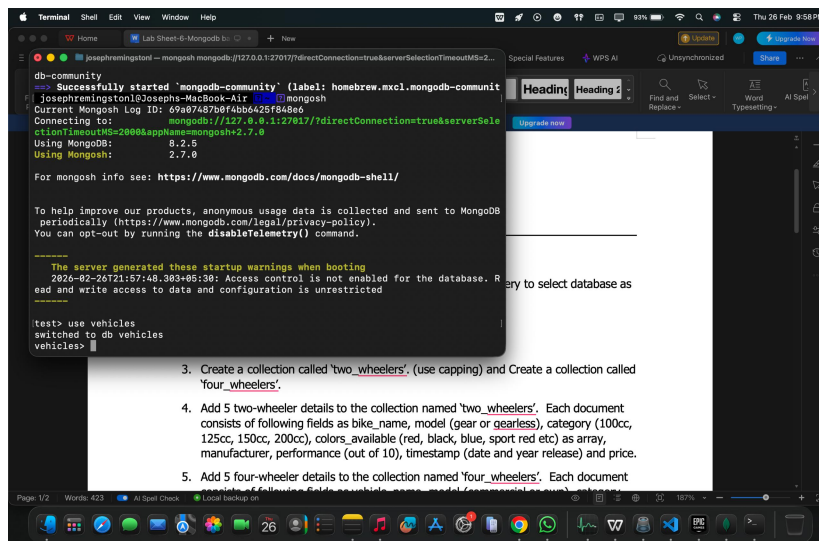
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Student name:

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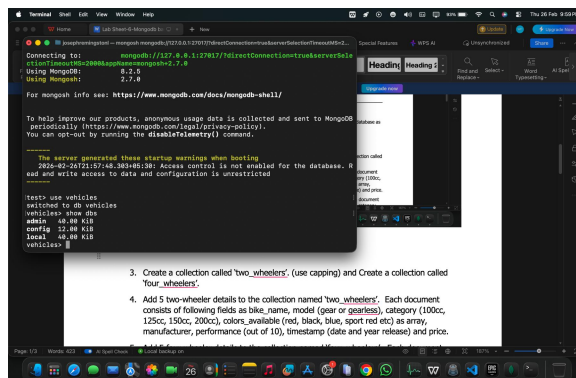
1. Use MongoDB to implement the following DB operations

1. Create a database called 'vehicles' and *write* a MongoDB query to select database as "vehicles".



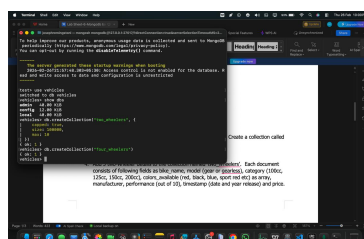
```
Terminal Shell Edit View Window Help
Lab Sheet 6-MongoDB bi...
Josephremington@Josephs-MacBook-Air:~$ mongosh
Successfully started 'mongodb-community' (label: homebrew.mxcl.mongodb-communit
Current Mongosh Log ID: 69a07487dbf4b6426f848e6
Connecting to: mongodb://127.0.0.1:27017/?directConnection=true&serverSele
Using MongoDB: 6.2.5
Using Mongosh: 2.7.0
For mongosh info see: https://www.mongodb.com/docs/mongosh-shell/
To help improve our products, anonymous usage data is collected and sent to MongoDB
periodically (https://www.mongodb.com/legal/privacy-policy).
You can opt-out by running the disableTelemetry() command.
The server generated these startup warnings when booting
2026-02-26T21:57:48.303+05:30: Access control is not enabled for the database. R
read and write access to data and configuration is unrestricted
(test> use vehicles
switched to db vehicles
vehicles>
```

2. Write a MongoDB query to display all the databases.



```
Terminal Shell Edit View Window Help
Lab Sheet 6-MongoDB bi...
Josephremington@Josephs-MacBook-Air:~$ mongosh
Successfully started 'mongodb-community' (label: homebrew.mxcl.mongodb-communit
Current Mongosh Log ID: 69a07487dbf4b6426f848e6
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2026-02-26T21:57:48.303+05:30: Access control is not enabled for the database. R
read and write access to data and configuration is unrestricted
(test> use vehicles
switched to db vehicles
vehicles> show dbs
admin 48.00 KiB
config 12.00 KiB
local 48.00 KiB
vehicles>
```

3. Create a collection called 'two_wheelers'. (use capping) and Create a collection called 'four_wheelers'.



```
Terminal Shell Edit View Window Help
Lab Sheet 6-MongoDB bi...
Josephremington@Josephs-MacBook-Air:~$ mongosh
Successfully started 'mongodb-community' (label: homebrew.mxcl.mongodb-communit
Current Mongosh Log ID: 69a07487dbf4b6426f848e6
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You can opt-out by running the disableTelemetry() command.
The server generated these startup warnings when booting
2026-02-26T21:57:48.303+05:30: Access control is not enabled for the database. R
read and write access to data and configuration is unrestricted
(test> use vehicles
switched to db vehicles
vehicles> use vehicles
vehicles> createCollection('two_wheelers')
{ok: true}
vehicles> createCollection('four_wheelers')
{ok: true}
vehicles>
```

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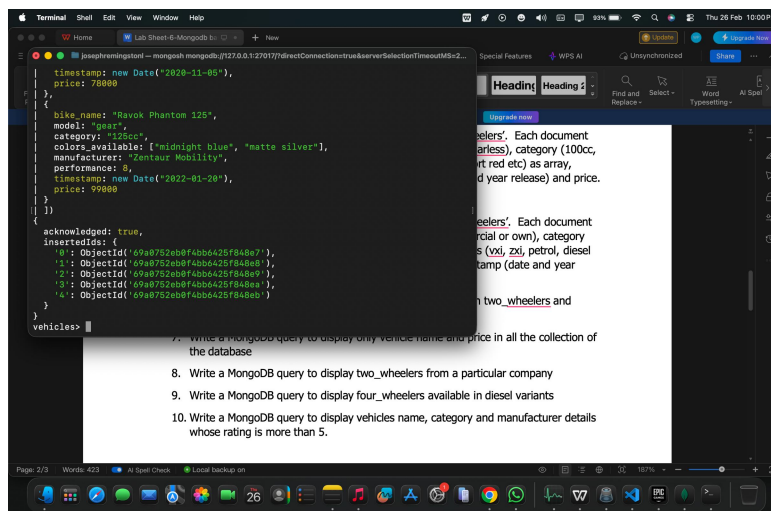
Faculty Name: Prof. S.Gopikrishnan

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4. Add 5 two-wheeler details to the collection named 'two_wheelers'. Each document consists of following fields as bike_name, model (gear or gearless), category (100cc, 125cc, 150cc, 200cc), colors_available (red, black, blue, sport red etc) as array, manufacturer, performance (out of 10), timestamp (date and year release) and price.



```

timestamp: new Date("2020-11-05"),
price: 78000
},
{
  bike_name: "Rayok Phantom 125",
  model: "gear",
  category: "125cc",
  colors_available: ["midnight blue", "matte silver"],
  manufacturer: "Zentaur Mobility",
  performance: 8,
  timestamp: new Date("2022-01-20"),
  price: 99000
}
})
acknowledged: true,
insertedIds: {
  0: ObjectId('69a8752dbf4bb6425f84be7'),
  1: ObjectId('69a8752dbf4bb6425f84be8'),
  2: ObjectId('69a8752dbf4bb6425f84be9'),
  3: ObjectId('69a8752dbf4bb6425f84bea'),
  4: ObjectId('69a8752dbf4bb6425f84beb')
}
}
vehicles>

```

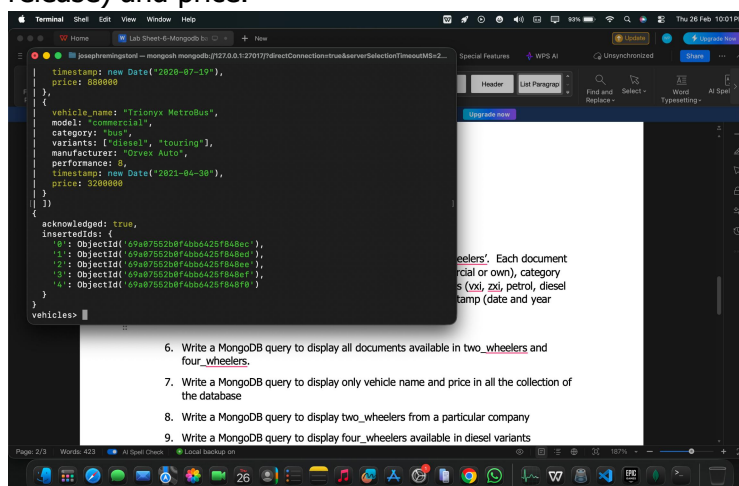
7. Write a MongoDB query to display only vehicle name and price in all the collection of the database

8. Write a MongoDB query to display two_wheelers from a particular company

9. Write a MongoDB query to display four_wheelers available in diesel variants

10. Write a MongoDB query to display vehicles name, category and manufacturer details whose rating is more than 5.

5. Add 5 four-wheeler details to the collection named 'four_wheelers'. Each document consists of following fields as vehicle_name, model (commercial or own), category (car, lorry, bus, mini truck, heavy truck, containers), variants (vxi, zxi, petrol, diesel etc) as array, manufacturer, performance (out of 10), timestamp (date and year release) and price.



```

timestamp: new Date("2020-07-19"),
price: 880000
},
{
  vehicle_name: "Trinixx MetroBus",
  model: "commercial",
  category: "bus",
  variants: ["diesel", "touring"],
  manufacturer: "Gover Auto",
  performance: 8,
  timestamp: new Date("2021-04-30"),
  price: 5200000
}
})
acknowledged: true,
insertedIds: {
  0: ObjectId('69a87552dbf4bb6425f84be7'),
  1: ObjectId('69a87552dbf4bb6425f84be8'),
  2: ObjectId('69a87552dbf4bb6425f84be9'),
  3: ObjectId('69a87552dbf4bb6425f84bea'),
  4: ObjectId('69a87552dbf4bb6425f84beb')
}
}
vehicles>

```

6. Write a MongoDB query to display all documents available in two_wheelers and four_wheelers.

7. Write a MongoDB query to display only vehicle name and price in all the collection of the database

8. Write a MongoDB query to display two_wheelers from a particular company

9. Write a MongoDB query to display four_wheelers available in diesel variants

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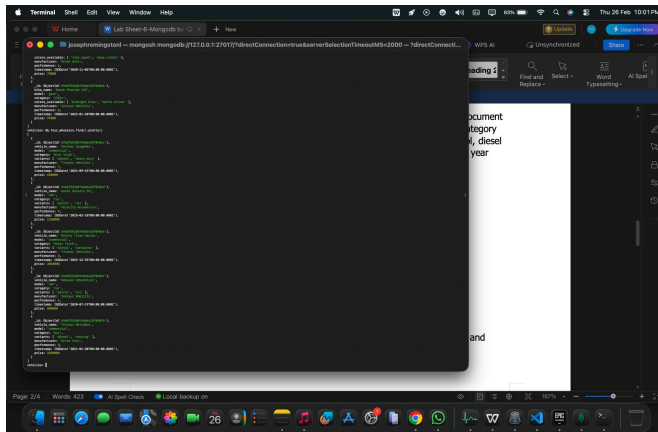
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Date:

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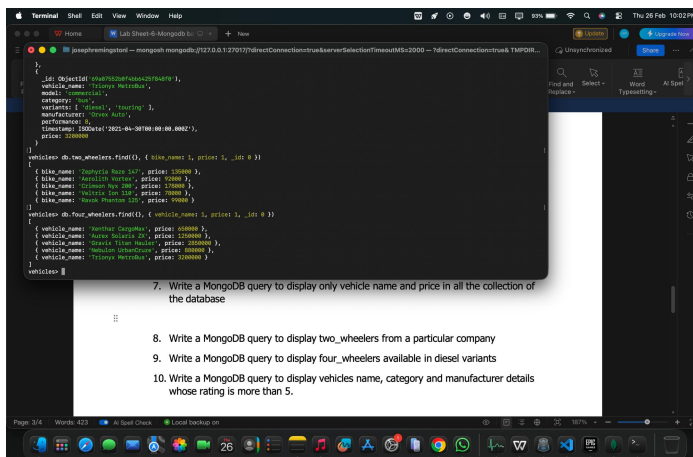
6. Write a MongoDB query to display all documents available in two_wheelers and four_wheelers.



```
use test;
show collections;
use test;
use two_wheelers;
find({})
```

document
category
year

7. Write a MongoDB query to display only vehicle name and price in all the collection of the database



```
use test;
show collections;
use test;
use two_wheelers;
find({}, {'_id': 0})

use test;
use four_wheelers;
find({}, {'_id': 0})
```

7. Write a MongoDB query to display only vehicle name and price in all the collection of the database

8. Write a MongoDB query to display two_wheelers from a particular company

9. Write a MongoDB query to display four_wheelers available in diesel variants

10. Write a MongoDB query to display vehicles name, category and manufacturer details whose rating is more than 5.

8. Write a MongoDB query to display two_wheelers from a particular company

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```
vehicles db:two_wheeler.find(
  { manufacturer: "Virexon Motors" }
)
{
  "_id": ObjectId("64b732baf4a42579b4e1"),
  "bike_name": "Supernova Neo 300",
  "model": "Neo300",
  "category": "Bike",
  "color_available": [ "matte blue", "ember red", "carbon black" ],
  "manufacturer": "Virexon Motors",
  "performance": 9,
  "timestamp": ISODate("2022-06-12T08:00:00.000Z"),
  "price": 230000
}
{
  "_id": ObjectId("64b732baf4a42579b4e2"),
  "bike_name": "Virexon Neo 200",
  "model": "Neo200",
  "category": "Bike",
  "color_available": [ "sport red", "onyx black" ],
  "manufacturer": "Virexon Motors",
  "performance": 9,
  "timestamp": ISODate("2023-06-01T08:00:00.000Z"),
  "price": 170000
}
vehicles
```

9. Write a MongoDB query to display four_wheeler available in diesel variants

```
vehicles db:four_wheeler.find(
  { variant: { $in: [ "diesel", "heavy duty" ] },
    manufacturer: "Virexon Motors",
    performance: 9,
    timestamp: ISODate("2021-09-25T08:00:00.000Z"),
    price: 600000
  }
)
{
  "_id": ObjectId("64b732baf4a42579b4e3"),
  "vehicle_name": "David Titan Reducer",
  "model": "TitanReducer",
  "category": "Heavy duty",
  "variant": "diesel",
  "manufacturer": "Virexon Motors",
  "performance": 9,
  "timestamp": ISODate("2022-12-25T08:00:00.000Z"),
  "price": 550000
}
{
  "_id": ObjectId("64b732baf4a42579b4e4"),
  "vehicle_name": "Virexon Motorbus",
  "model": "Motorbus",
  "category": "Bus",
  "variant": "diesel",
  "manufacturer": "Virexon Motors",
  "performance": 9,
  "timestamp": ISODate("2021-06-30T08:00:00.000Z"),
  "price": 500000
}
vehicles
```

9. Write a MongoDB query to display four_wheeler available in diesel variants

10. Write a MongoDB query to display vehicles name, category and manufacturer details whose rating is more than 5.

10. Write a MongoDB query to display vehicles name, category and manufacturer details whose rating is more than 5.

```
vehicles db:four_wheeler.find(
  { performance: { $gt: 5 } }
)
{
  "_id": ObjectId("64b732baf4a42579b4e3"),
  "vehicle_name": "David Titan Reducer",
  "model": "TitanReducer",
  "category": "Heavy duty",
  "variant": "diesel",
  "manufacturer": "Virexon Motors",
  "performance": 9,
  "timestamp": ISODate("2022-12-25T08:00:00.000Z"),
  "price": 550000
}
{
  "_id": ObjectId("64b732baf4a42579b4e4"),
  "vehicle_name": "Virexon Motorbus",
  "model": "Motorbus",
  "category": "Bus",
  "variant": "diesel",
  "manufacturer": "Virexon Motors",
  "performance": 9,
  "timestamp": ISODate("2021-06-30T08:00:00.000Z"),
  "price": 500000
}
vehicles
```

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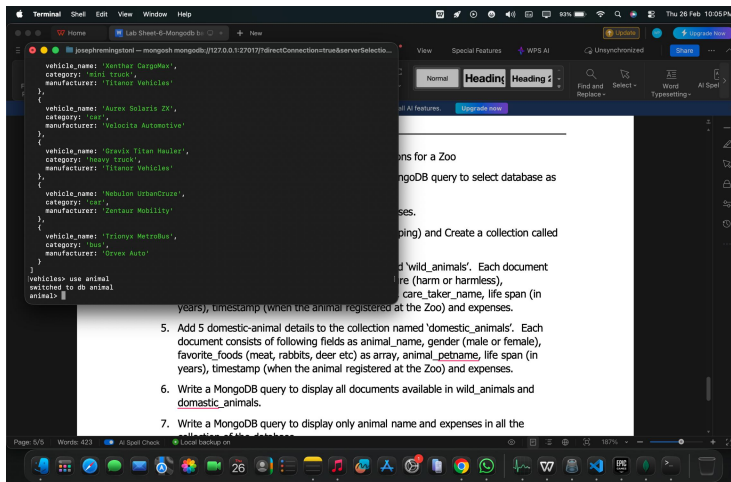
Date:

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Reg. no.:

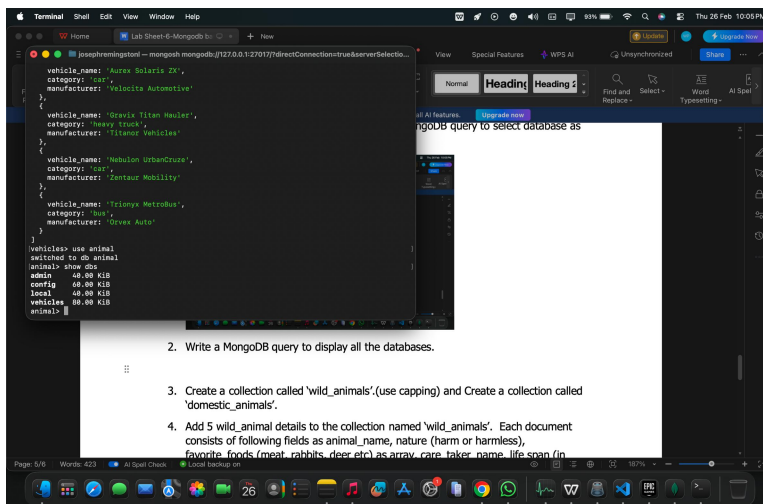
2. Use MongoDB to implement the following DB operations for a Zoo

1. Create a database called 'animal' and write a MongoDB query to select database as 'animal'.



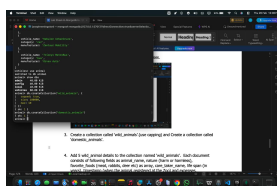
```
use animal
createDatabase('animal')
use animal
createCollection('vehicles')
insert('vehicles', { 'vehicle_name': 'Xanthar CargoBox', 'category': 'semi truck', 'manufacturer': 'Titanor Vehicles' }, { 'vehicle_name': 'Aurex Solaris ZX', 'category': 'car', 'manufacturer': 'Velocita Automotive' }, { 'vehicle_name': 'Gravis Titan Hauler', 'category': 'heavy truck', 'manufacturer': 'Titanor Vehicles' }, { 'vehicle_name': 'Nebulon UrbanCruze', 'category': 'bus', 'manufacturer': 'Zentaur Mobility' }, { 'vehicle_name': 'Trionyx Metrobus', 'category': 'bus', 'manufacturer': 'Orevox Auto' })
show collections
show databases
```

2. Write a MongoDB query to display all the databases.



```
use animal
show databases
```

3. Create a collection called 'wild_animals'.(use capping) and Create a collection called 'domestic_animals'.



```
use animal
createCollection('wild_animals')
createCollection('domestic_animals')
```

Date:

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-
- The screenshot shows a Mac desktop with a terminal window and a MongoDB Compass interface. The terminal window displays a MongoDB query that inserts a document into the 'wild_animals' collection. The document contains fields for 'nature', 'favorite_foods', 'care_taker_name', 'life_span', 'expenses', 'animal_name', and 'timestamp'. The MongoDB Compass interface shows the 'wild_animals' collection with a document that has similar fields, including 'name', 'gender', 'pet_name', and 'expenses'.
- ```

{
 "nature": "harmless",
 "favorite_foods": ["grass", "shrubs"],
 "care_taker_name": "Hestia D.",
 "life_span": 12,
 "timestamp": new Date("2023-01-05"),
 "expenses": 10000
},
{
 "animal_name": "Golden Dune Viper",
 "nature": "poison",
 "favorite_foods": ["rats", "lizards"],
 "care_taker_name": "Aditi Rao",
 "life_span": 6,
 "timestamp": new Date("2021-11-20"),
 "expenses": 18000
},
{
 "acknowledged": true,
 "insertedIds": [
 "659b76ac0bf4b06a42f5f848f71",
 "659b76ac0bf4b06a42f5f848f72",
 "659b76ac0bf4b06a42f5f848f73",
 "659b76ac0bf4b06a42f5f848f74",
 "659b76ac0bf4b06a42f5f848f75"
]
}
}

```
6. Write a MongoDB query to display all documents available in wild\_animals and domestic\_animals.
7. Write a MongoDB query to display only animal name and expenses in all the collection of the database
8. Write a MongoDB query to display domestic\_animals whose life is a particular year
9. Write a MongoDB query to display wild\_animals available under a particular care\_taker

- 
- The screenshot shows a MongoDB Shell window with a JSON document for an animal named 'Pygmy Goat'. The document is as follows:
- ```
{
  "gender": "male",
  "favorite_foods": ["seeds", "fruits"],
  "animal_petname": "Kali",
  "life_span": 7,
  "timestamp": new Date("2023-05-22"),
  "expenses": 6000
},
{
  "animal_name": "Pygmy Goat",
  "gender": "female",
  "favorite_foods": ["grass", "grains"],
  "animal_petname": "Bella",
  "life_span": 10,
  "timestamp": new Date("2022-08-30"),
  "expenses": 15000
},
{
  "acknowledged": true,
  "insertedIds": [
    '659b76c9b8f4bb6425f8a876',
    '659b76c9b8f4bb6425f8a877',
    '659b76c9b8f4bb6425f8a878',
    '659b76c9b8f4bb6425f8a879',
    '659b76c9b8f4bb6425f8a87a'
  ]
}
```
- The document is part of a collection named 'domestic_animals'. The text on the right side of the image indicates that the collection contains documents for 'domestic_animals', each with fields for name, gender (male or female), animal_petname, life span (in years), and expenses.

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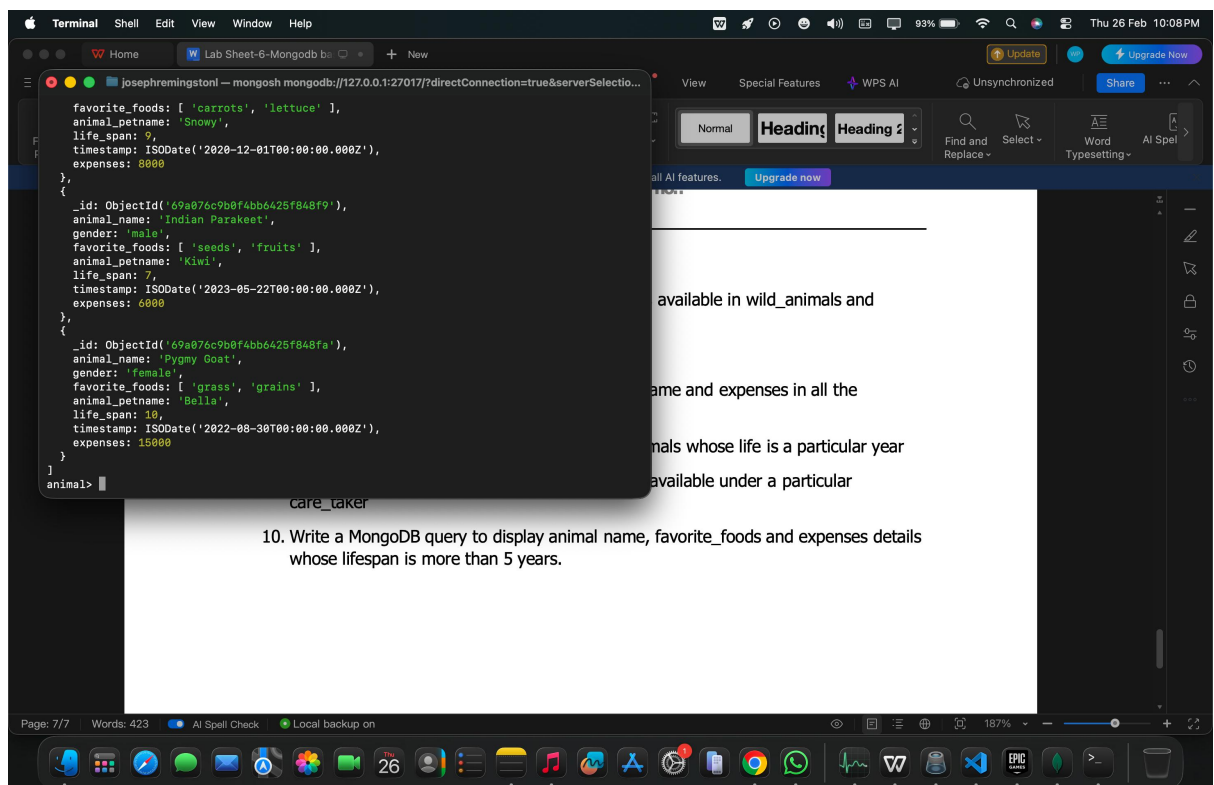
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Date:

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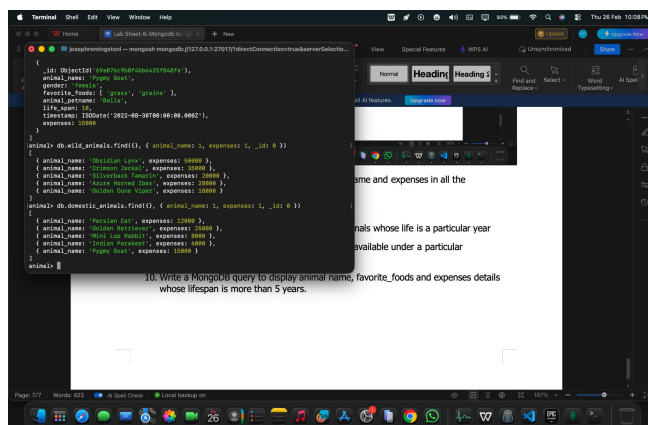
Reg. no.:

6. Write a MongoDB query to display all documents available in wild_animals and domestic_animals.



10. Write a MongoDB query to display animal name, favorite_foods and expenses details whose lifespan is more than 5 years.

7. Write a MongoDB query to display only animal name and expenses in all the collection of the database



10. Write a MongoDB query to display animal name, favorite_foods and expenses details whose lifespan is more than 5 years.

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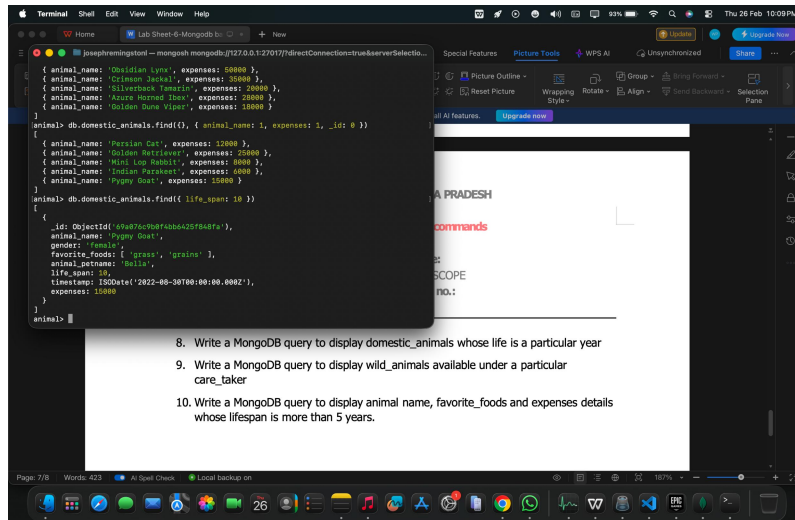
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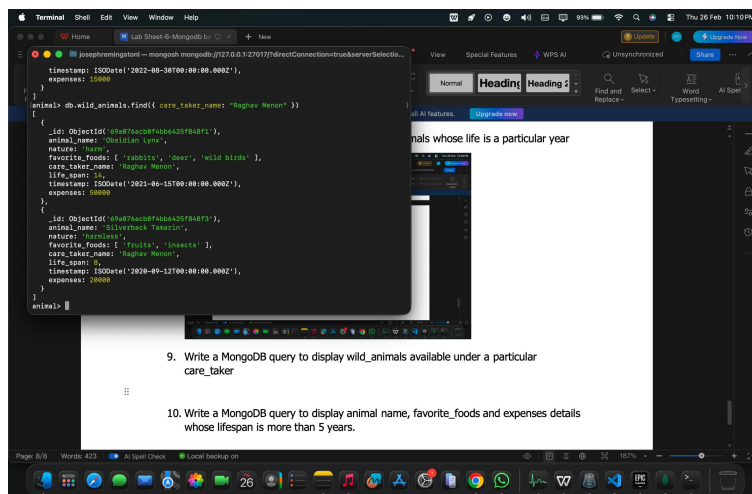
Reg. no.:

8. Write a MongoDB query to display domestic_animals whose life is a particular year



```
terminal> mongo --host localhost --port 27017 --eval 'use animals'
terminal> db.domestic_animals.find( { animal_name: 1, expenses: 1, _id: 0 } )
{
  "_id" : ObjectId("69a876c0b4f4bb425f84fa"),
  "animal_name" : "Persian Cat",
  "expenses" : 12000,
  "life_span" : 15,
  "timestamp" : ISODate("2022-08-30T00:00:00.000Z"),
  "expenses" : 12000
}
terminal> db.domestic_animals.find( { life_span: 10 } )
{
  "_id" : ObjectId("69a876c0b4f4bb425f84fa"),
  "animal_name" : "Pygmy Goat",
  "expenses" : 12000,
  "life_span" : 10,
  "timestamp" : ISODate("2022-08-30T00:00:00.000Z"),
  "expenses" : 12000
}
```

9. Write a MongoDB query to display wild_animals available under a particular care_taker



```
terminal> db.wild_animals.find( { care_taker_name: "Raghav Menon" } )
{
  "_id" : ObjectId("69a876c0b4f4bb425f84fa"),
  "animal_name" : "Persian Cat",
  "expenses" : 12000,
  "life_span" : 15,
  "timestamp" : ISODate("2022-08-30T00:00:00.000Z"),
  "expenses" : 12000
}
terminal> db.wild_animals.find( { care_taker_name: "Raghav Menon" } )
{
  "_id" : ObjectId("69a876c0b4f4bb425f84fa"),
  "animal_name" : "Pygmy Goat",
  "expenses" : 12000,
  "life_span" : 10,
  "timestamp" : ISODate("2022-08-30T00:00:00.000Z"),
  "expenses" : 12000
}
```

10. Write a MongoDB query to display animal name, favorite_foods and expenses details whose lifespan is more than 5 years.

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The screenshot shows a macOS desktop environment. In the foreground, a WPS AI editor window is open, displaying a question: "10. Write a MongoDB query to display animal name, favorite_foods and expenses details whose lifespan is more than 5 years." Below the question, there are three vertical ellipsis dots. In the background, a terminal window is open, showing a MongoDB shell session. The terminal displays a JSON array of documents representing animal records. The documents are as follows:

```
[
  {
    animal_name: 'Persian Cat',
    favorite_foods: [ 'fish', 'chicken' ],
    expenses: 12000
  },
  {
    animal_name: 'Golden Retriever',
    favorite_foods: [ 'meat', 'dog food' ],
    expenses: 25000
  },
  {
    animal_name: 'Mini Lop Rabbit',
    favorite_foods: [ 'carrots', 'lettuce' ],
    expenses: 8000
  },
  {
    animal_name: 'Indian Parakeet',
    favorite_foods: [ 'seeds', 'fruits' ],
    expenses: 6000
  },
  {
    animal_name: 'Pygmy Goat',
    favorite_foods: [ 'grass', 'grains' ],
    expenses: 15000
  }
]
```

The terminal prompt is `animal>`. The WPS AI editor window has a top bar with "Update" and "Upgrade Now" buttons, and a right sidebar with "Find and Replace", "Select", "Word", and "AI Spell" options. The bottom status bar of the WPS AI editor shows "Page: 8/8", "Words: 423", "AI Spell Check", and "Local backup on". The macOS dock at the bottom contains various application icons, including Safari, Mail, Messages, Photos, App Store, Calendar, Reminders, Notes, Music, Podcasts, TV, Health, Fitness, Weather, Clock, Calculator, and System Preferences.